Topic 4 — Light, Optics and Laser Technology

Direct Syllabus Mapping: General Science — Optics (Nature of Light, Wave-Particle Duality, Speed in Different Media, Rectilinear Propagation, Reflection, Refraction, Snell's Law, Total Internal Reflection & Applications, Optical Fibers, Endoscopes, Dispersion, Prisms, Rainbow Formation, Scattering, Rayleigh's Law, Mirrors, Lenses, Lens Power in Dioptres, Human Eye Anatomy & Defects, Astigmatism, Presbyopia, Double-slit Interference, Polarization, Primary & Secondary Colors, Color TV, Laser Technology).
Key Must-Score Focus: Plane mirror focal length = ∞ (71st BPSC 2025), Rectilinear propagation forming shadows (70th BPSC Re-Exam 2024), Telescope objective diameter & resolution (70th BPSC Re-Exam 2024), Young's double-slit fringe width $\beta = 0.5\ \text{mm}$ (UKPCS 2024), Non-luminous Moon (UP RO/ARO 2023), Concave mirror forming same size image at C (69th BPSC 2023), Thin black strips on lens reducing brightness (68th BPSC 2022), Soap film interference (67th BPSC 2022), Twinkling of stars via atmospheric refraction (67th BPSC 2022), Eye sensitivity to yellowish-green $555\ \text{nm}$ (UP RO/ARO 2021), Albedo of fresh snow $80 - 85\%$ (UPPCS 2021), Critical angle of diamond 24.4°, Air bubble in water behaving as concave/divergent lens, Minimum plane mirror height $= H/2$, and Color mixing matrices (RGB vs CMY).

Current Affairs

- **71st BPSC (Pre 2025) Optics Fact**:
- *Focal Length of Plane Mirror*: Treated as the limiting case of a spherical mirror with an infinite radius of curvature ($R = \infty$) $\implies$ **Focal length $f = R/2 = \infty$ (Infinity)**. Its optical power is $P = 1/f = 0\ \text{Dioptre}$.
- **70th BPSC (Pre Re-Exam 2024) Dual Facts**:
- *Shadow Formation*: Shadows are produced directly due to the **rectilinear propagation of light** (light travels in straight lines and cannot bend around macroscopic opaque objects).
- *Telescope Objective Diameter & Resolution*: As the aperture diameter (D) of a telescope's objective lens increases, diffraction decreases and light-gathering power increases $\implies$ **Resolving power / resolution increases** (Rayleigh criterion $\theta = 1.22\lambda/D$).
- **UKPCS (Pre 2024) Young's Double-Slit Numerical**:
- Wavelength $\lambda = 500\ \text{nm} = 5 \times 10^{-7}\ \text{m}$, screen distance $D = 1\ \text{m}$, slit separation $d = 1\ \text{mm} = 10^{-3}\ \text{m}$.
- Fringe separation (fringe width): $\beta = \frac{\lambda D}{d} = \frac{5 \times 10^{-7} \times 1}{10^{-3}} = 5 \times 10^{-4}\ \text{m} = \mathbf{0.5\ mm}$.
- **69th BPSC (Pre 2023) Colour Combination Fact**:
- Primary pigments & complementary colors: **Magenta = Red + Blue**; **Cyan = Green + Blue**; **Teal = Blue + Green + White**; **Mauve = Blue + Red + White**.
- **68th BPSC (Pre 2022) Lens Intensity Fact**:
- If thin strips of black paint are drawn on a convex lens catching the image of a white horse, it produces an **image of a horse with reduced brightness** (the strips block a fraction of incident rays without casting localized shadows).

- **Human Eye Physiological Response (UP RO/ARO 2021 Fact)**:
- The human eye is most sensitive to **yellowish-green light of wavelength** $\approx 555\ \text{nm}$ **(**$5550\ \text{\AA}$**)** under photopic (daylight) conditions.
- Human ear sensitivity peaks between $1000\ \text{Hz}$ **and** $5000\ \text{Hz}$.
- **Persistence of Vision** is $1/16\text{th}$ **of a second**; **Persistence of Hearing** is $1/10\text{th}$ **of a second**.
- **Albedo & Fresh Snow (UPPCS Pre 2021)**:
- Fresh snow and clean ice have the highest albedo on Earth, reflecting 80% **to** 85% of incident solar radiation.

Teaching Cards — Light, Optics and Laser

4.1 Nature of Light

Light is electromagnetic radiation that the human eye can detect, roughly from $380\ \text{nm}$ **to** $750\ \text{nm}$.

- In everyday optics, light is treated as travelling in straight **rays** in a homogeneous medium.
- **Rectilinear propagation** means light travels in straight lines and forms sharp shadows behind opaque objects.
- The **wave model** explains interference, diffraction, and polarisation.
- The **particle (photon) model** explains the photoelectric effect and discrete energy exchange.
- Light shows **wave–particle duality**: different phenomena need the ray, wave, or photon picture.
- In vacuum, the speed of light is maximum: $c = 3.0 \times 10^8\ \text{m/s}$.
- Frequency of light does **not** change when light enters a new medium.
- Speed and wavelength both change in a new medium according to $v = f\lambda$.
- Speed of light is smaller in denser optical media: vacuum $>$ water $>$ glass $>$ diamond.
- A **luminous** body emits its own light.
- The **Moon** is non-luminous and is seen only by reflected sunlight.
- **Albedo** is the fraction of incident radiation that a surface reflects.
- Fresh snow has a very high albedo of about 80% **to** 85%.

4.2 Reflection of Light

Reflection is the return of light into the same medium after striking a surface.

- The **incident ray**, the **reflected ray**, and the **normal** at the point of incidence all lie in one plane.
- The **angle of incidence** equals the **angle of reflection** ($i = r$).
- A **plane mirror** forms a virtual, erect, laterally inverted image of the same size as the object.
- The image in a plane mirror lies as far behind the mirror as the object lies in front.
- Focal length of a plane mirror is treated as **infinity**, so its power is **zero**.
- If a plane mirror rotates by angle θ, the reflected ray rotates by 2θ.
- The minimum vertical height of a plane mirror to see one's full standing height H is $H/2$.

- Two parallel plane mirrors form an **infinite** number of images of an object between them.
- For two mirrors inclined at angle θ, the number of images is $n = \frac{360^\circ}{\theta} - 1$ when $360/\theta$ is even.
- A **concave mirror** has its reflecting surface curved inward and acts as a converging mirror.
- A concave mirror can form real inverted images for objects beyond the focus.
- When the object is between the pole and the focus of a concave mirror, the image is **virtual, erect, and magnified**.
- When the object is at the centre of curvature C of a concave mirror, the image is at C, real, inverted, and of the **same size**.
- Concave mirrors are used in shaving mirrors, dentist mirrors, car headlights, searchlights, and solar furnaces.
- In a car headlight, the bulb is placed at the focus so reflected rays leave as a nearly **parallel beam**.
- A **convex mirror** has its reflecting surface curved outward and always acts as a diverging mirror.
- A convex mirror always forms a **virtual, erect, and diminished** image behind the mirror.
- Vehicle **rear-view and side mirrors** use convex mirrors because they give a wide field of view.

4.3 Refraction of Light

Refraction is the bending of light when it crosses the boundary between two media of different optical density.

- Light slows down and bends **towards the normal** when it enters an optically denser medium.
- Light speeds up and bends **away from the normal** when it enters an optically rarer medium.
- **Snell's law** states that $\mu_1 \sin i = \mu_2 \sin r$, or $\frac{\sin i}{\sin r} = {}_1\mu_2$.
- Absolute refractive index of a medium is $\mu = \frac{c}{v}$, where v is the speed of light in that medium.
- Absolute refractive index of water is about $4/3 \approx 1.33$.
- Absolute refractive index of diamond is about 2.42, the highest among common transparent solids.
- Apparent depth of a pool is less than real depth because rays from the bottom bend away from the normal on leaving water.
- Real depth and apparent depth are related by $\text{real depth} = \mu \times \text{apparent depth}$.
- A stick partly immersed in water appears bent at the water surface due to refraction.
- A coin in a bucket of water appears raised due to refraction.
- An underwater diver looking upward sees aerial objects at a **higher** apparent level than their true height.
- **Twinkling of stars** is caused by continuous atmospheric refraction through moving air layers of changing density.
- Atmospheric refraction makes the Sun visible about 2 **minutes** before geometric sunrise and about 2 **minutes** after geometric sunset.
- Near the horizon, unequal refraction across the disc makes the Sun or Moon look **elliptical**.

4.4 Total Internal Reflection (TIR)

Total internal reflection is complete reflection of light at a denser–rarer boundary when the angle of incidence exceeds the critical angle.

- For TIR, light must travel from an optically **denser** medium toward a **rarer** medium.
- The angle of incidence must be greater than the **critical angle** i_c.
- Critical angle satisfies $\sin i_c = \frac{1}{\mu}$ for a denser medium against air.
- Critical angle for water–air is about 48.75°.
- Critical angle for glass–air is about 41°.
- Critical angle for diamond–air is about 24.4°, which is unusually small.
- A cut diamond sparkles because repeated TIR traps and redirects light inside the gem.
- An **optical fibre** has a core of higher refractive index surrounded by cladding of lower refractive index.
- Light travels along an optical fibre by repeated total internal reflection with very small loss.
- Optical fibres are used in high-speed telecommunication because they resist electromagnetic interference.
- An **endoscope** uses flexible fibre bundles and TIR to view internal organs.
- A desert **mirage** forms when light from the sky undergoes TIR in hot, optically rarer air near the ground.
- Tiny glass beads on road signs and number plates return headlight beams by TIR and refraction.

4.5 Lenses, Power and the Human Eye

A lens is a transparent refracting device with two curved surfaces that converges or diverges light.

- A **convex (converging) lens** is thicker at the centre than at the edges and has positive focal length.
- A **concave (diverging) lens** is thinner at the centre than at the edges and has negative focal length.
- Lens power is $P = \frac{1}{f}$, with f in metres, and the unit is the **dioptre (D)**.
- A lens of power $+2\ \mathrm{D}$ has focal length $f = 0.50\ \mathrm{m} = 50\ \mathrm{cm}$.
- Convex lenses are used as magnifying glasses, in cameras, microscopes, and telescopes.
- When an object lies between the optical centre and the focus of a convex lens, the image is virtual, erect, and magnified.
- A concave lens always forms a virtual, erect, and diminished image.
- An **air bubble in water** has convex shape but behaves as a **concave (diverging) lens**.
- If a convex lens and a concave lens of equal focal length are placed in contact, net power is zero and the pair acts like a plane glass sheet.
- The least distance of distinct vision for a normal adult eye is $25\ \mathrm{cm}$.
- The eye lens changes curvature by action of the **ciliary muscles** during accommodation.
- The image on the retina is real, inverted, and diminished.
- **Myopia (near-sightedness)** forms the image of a distant object **in front of** the retina.
- Myopia is corrected by a **concave (diverging) lens**.
- **Hypermetropia (far-sightedness)** forms the image of a near object **behind** the retina.
- Hypermetropia is corrected by a **convex (converging) lens**.
- **Presbyopia** is age-related loss of near focusing and is corrected by a bifocal or progressive lens.
- **Astigmatism** comes from uneven corneal or lens curvature and is corrected by a **cylindrical lens**.

- **Cataract** is clouding of the crystalline lens and is treated by surgical replacement with an intraocular lens.
- The human eye under daylight vision is most sensitive near **yellowish-green light of about** $555\ \text{nm}$.

4.6 Prism, Dispersion and Rainbow

Dispersion is the splitting of white light into its constituent colours because refractive index depends on wavelength.

- A glass prism refracts white light twice and spreads it into the VIBGYOR spectrum.
- **Violet** has the shortest visible wavelength, highest frequency in glass, highest refractive index, and the **largest** deviation.
- **Red** has the longest visible wavelength, lowest frequency in glass, lowest refractive index, and the **smallest** deviation.
- Primary rainbow colours appear with **red outside** and **violet inside**.
- Green lies in the middle of the primary rainbow band.
- A primary rainbow forms by **two refractions and one total internal reflection** inside a raindrop, together with dispersion.
- The observer sees a rainbow in the direction **opposite** to the Sun.
- A natural rainbow cannot be seen at **local noon** when the Sun is nearly overhead.
- Shimmering colours on a soap film or oil slick are due to **interference**, not prism dispersion.
- Rainbow colours on a CD in sunlight come from **reflection and diffraction** at the track grating.

4.7 Scattering of Light

Scattering is the redirection of light by particles or molecules that are small compared with the wavelength.

- **Rayleigh's law** states that scattered intensity varies as $I \propto \frac{1}{\lambda^4}$.
- Shorter wavelengths such as blue and violet scatter much more than red.
- The clear daytime sky looks **blue** because air molecules scatter short-wavelength blue light toward the observer.
- At sunrise and sunset, sunlight travels a longer atmospheric path, so blue light is scattered away and the disc looks **red or orange**.
- At very high altitude or in space with almost no atmosphere, the sky looks **black**.
- Danger and stop signals use **red** light because red suffers the least atmospheric scattering and travels farthest through fog and smoke.
- Green leaves look green because chlorophyll **reflects** green wavelengths and absorbs much of the rest of the visible band.
- The **Raman effect** is inelastic scattering of photons in a transparent medium and was discovered by C.V. Raman.

4.8 Laser

LASER means Light Amplification by Stimulated Emission of Radiation.

- Laser light is highly **monochromatic**, meaning it has a very narrow wavelength spread.
- Laser light is **coherent**, meaning the waves stay in a fixed phase relationship.
- Laser light is highly **directional (unidirectional)** and spreads far less than ordinary lamp light.
- Laser beams can carry high intensity in a narrow cone and are used for cutting, welding, and drilling.
- Semiconductor lasers read microscopic pits on CDs, DVDs, and Blu-ray discs.
- Medical uses of lasers include eye surgery, tissue cutting, and precise coagulation.
- Holography records and reconstructs a true three-dimensional image using coherent laser interference.
- Ordinary white light is broadband, incoherent, and spreads in many directions, unlike laser light.

4.9 Photoelectric Effect and Photovoltaic Conversion

Photoelectric and photovoltaic effects convert light energy into electrical energy or free electrons.

- In the **photoelectric effect**, photons eject electrons from a metal surface when their energy exceeds the work function.
- Photon energy is $E = h\nu = \frac{hc}{\lambda}$, so shorter wavelength light carries more energy per photon.
- Below a threshold frequency, increasing intensity alone does **not** start electron emission.
- Above threshold, the number of emitted electrons rises with light intensity, while maximum kinetic energy depends on frequency.
- A **photovoltaic (solar) cell** uses a semiconductor junction to convert sunlight directly into electric current.
- Solar panels generate DC electricity by the photovoltaic effect in silicon or related semiconductors.
- Photocells, light meters, and automatic street-light sensors use photoelectric or related light-to-electric conversion.
- Photosynthesis also stores light energy chemically, but photovoltaic cells produce usable electrical power directly.

Master Tables & Core Optical Formulations

Table 1: Master Summary of Image Formation by Spherical Mirrors

Mirror Type	Position of Object	Position of Image	Nature of Image	Relative Size
Concave	At Infinity	At Focus (F)	Real and Inverted	Highly Diminished (Point)
Concave	Beyond Centre of Curvature (C)	Between F and C	Real and Inverted	Diminished

Mirror Type	Position of Object	Position of Image	Nature of Image	Relative Size
Concave	At Centre of Curvature (C)	At C	Real and Inverted	**Same Size as Object**
Concave	Between C and F	Beyond C	Real and Inverted	Magnified
Concave	At Focus (F)	At Infinity	Real and Inverted	Highly Magnified
Concave	Between Pole (P) and Focus (F)	Behind the Mirror	**Virtual and Erect**	**Magnified (Shaving / Dentist)**
Convex	Anywhere between ∞ and Pole (P)	Behind mirror (between P & F)	**Virtual and Erect**	**Diminished (Rear-View Mirror)**

Table 2: Human Eye Defects, Optical Causes and Ophthalmic Corrections

Eye Defect	Common Name	Nature of Vision Impairment	Optical Cause	Corrective Lens
Myopia	Near-sightedness	Close objects clear; distant objects blurred	Eyeball too long or cornea curvature too sharp; image forms **in front of retina**	**Concave (Diverging) Lens** (Negative power)
Hypermetropia	Far-sightedness	Distant objects clear; close objects blurred	Eyeball too short or lens curvature too flat; image forms **behind retina**	**Convex (Converging) Lens** (Positive power)
Presbyopia	Old-age vision	Inability to focus on near objects with aging	Loss of elasticity of crystalline lens and weakening of ciliary muscles	**Bifocal Lens** (Upper concave for distance, lower convex for reading)
Astigmatism	Astigmatism	Distorted or blurred vision in all directions	Cornea or lens has irregular curvature (aspherical)	**Cylindrical Lens**
Cataract	Cataract	Progressive clouding of vision	Opaque crystalline protein aggregation in natural lens	Surgical lens extraction & Intraocular Lens (IOL) implant

Table 3: Refractive Indices, Speed of Light & Critical Angles Matrix

Medium	Absolute Refractive Index (μ)	Speed of Light in Medium (m/s)	Critical Angle for Total Internal Reflection (i_c)
Vacuum	1.0000 (Standard)	3.00×10^8	—

Medium	Absolute Refractive Index ( )	Speed of Light in Medium ( )	Critical Angle for Total Internal Reflection ( )
Air	1.0003	$\approx 3.00 \times 10^8$	—
Water	1.333 $(4/3)$	2.25×10^8	48.75°
Crown Glass	1.52	1.97×10^8	41.14°
Flint Glass	1.66	1.81×10^8	37.04°
Diamond	2.42 (Highest)	1.24×10^8 (Lowest)	24.41° (Lowest)

Table 4: Master Colorimetry & Light Spectrum Matrix

Color Band	Wavelength Range (nm)	Wavelength Range ($\AA$)	Deviation in Prism	Scattering Propensity	Energy per Photon (eV)
Violet	$380 - 450$	$3800 - 4500$	**Maximum**	**Maximum** ($I \propto 1/\lambda^4$)	High ($3.10 - 2.75$ eV)
Indigo	$450 - 485$	$4500 - 4850$	Very High	Very High	$2.75 - 2.56$ eV
Blue	$485 - 500$	$4850 - 5000$	High	High (Gives sky blue color)	$2.56 - 2.48$ eV
Green	$500 - 565$	$5000 - 5650$	Moderate	Moderate	$2.48 - 2.19$ eV
Yellow	$565 - 590$	$5650 - 5900$	Moderate-Low	Moderate-Low	$2.19 - 2.10$ eV
Orange	$590 - 625$	$5900 - 6250$	Low	Low	$2.10 - 1.98$ eV
Red	$625 - 750$	$6250 - 7500$	**Minimum**	**Minimum** (Used in danger signals)	Low ($1.98 - 1.65$ eV)

Bilingual Terminology -- Light, Optics and Laser

English Term	Hindi Term	Key Anchor
Light (प्रकाश)	प्रकाश	Electromagnetic radiation visible to human eye; 400-700 nm
Reflection (परावर्तन)	परावर्तन	Bouncing back of light from a surface
Refraction (अपवर्तन)	अपवर्तन	Bending of light when passing from one medium to another
Refractive Index (अपवर्तनांक)	अपवर्तनांक	n = speed of light in vacuum / speed of light in medium

English Term	Hindi Term	Key Anchor
Total Internal Reflection (पूर्ण आन्तरिक परावर्तन)	पूर्ण आन्तरिक परावर्तन	Light reflects back inside denser medium; basis of optical fibre
Lens (लेंस)	लेंस	Transparent object that refracts light; converging (convex) / diverging (concave)
Mirror (दर्पण)	दर्पण	Smooth reflective surface; plane, concave (converging), convex (diverging)
Dispersion (प्रकाश का प्रकीर्णन)	प्रकाश का वर्ण-विक्षेपण	Splitting of white light into VIBGYOR spectrum (by prism)
VIBGYOR	बैनीआहपीनाला	Violet, Indigo, Blue, Green, Yellow, Orange, Red
Scattering (प्रकाश प्रकीर्णन)	प्रकीर्णन	Redirection of light by particles; Rayleigh scattering (sky is blue)
Optical Fibre (प्रकाशिक तंतु)	प्रकाशिक तंतु	Carries light by total internal reflection; used in telecom, endoscopy
LASER (लेजर)	लेज़र	Light Amplification by Stimulated Emission of Radiation; monochromatic, coherent
Myopia (निकट दृष्टि दोष)	निकट दृष्टि / मायोपिया	Cannot see far; corrected by **concave lens**
Hypermetropia (दूर दृष्टि दोष)	दूर दृष्टि / हाइपरमेट्रोपिया	Cannot see near; corrected by **convex lens**
Astigmatism (दृष्टिवैषम्य)	दृष्टिवैषम्य	Uneven curvature of cornea; blurred vision; cylindrical lens
Presbyopia (जरा दूर दृष्टिता)	जरा दूर दृष्टि	Age-related vision problem; bifocal lens (invented by Benjamin Franklin)

One-Liner Revision -- Light, Optics and Laser

#	Fast Fact
1	Speed of light in vacuum (c) = **3 x 10^8 m/s** (299,792,458 m/s exactly)
2	Light is both wave (interference, diffraction) and particle (photoelectric effect)
3	Visible spectrum = 400-700 nm; violet (400nm) to red (700nm)
4	VIBGYOR spectrum: Violet has shortest wavelength; Red has longest wavelength
5	Speed of light = lowest in **diamond** (n=2.42; highest refractive index)
6	Laws of reflection: angle of incidence = angle of reflection; in same plane
7	Concave mirror = **converging** mirror; used in torches, telescopes, solar furnaces

#	Fast Fact
8	Convex mirror = **diverging** mirror; used in rear-view mirrors, security mirrors
9	Convex lens = **converging**; used in magnifying glass, camera, eye
10	Concave lens = **diverging**; corrects myopia (short-sightedness)
11	Myopia = cannot see far; eyeball too long; corrected by **concave lens**
12	Hypermetropia = cannot see near; eyeball too short; corrected by **convex lens**
13	Presbyopia = old-age vision; bifocals (convex for near, concave for far)
14	Total Internal Reflection occurs when light goes from **denser to rarer** medium above critical angle
15	Diamond sparkles due to **total internal reflection** (critical angle = 24.4 deg)
16	Optical fibre uses TIR to transmit light signals; glass/silica core
17	Sky is blue = **Rayleigh scattering**; blue light scattered more (shorter wavelength)
18	Sunset/sunrise red = blue scattered away; long-wavelength red remains
19	Mirages = due to total internal reflection of light in hot air layers near ground
20	Rainbow = formed by reflection + refraction + dispersion in water droplets
21	Dispersion: violet light bends most; red light bends least (through prism)
22	Primary colours of light = Red, Green, Blue (RGB) -- additive mixing
23	Primary colours of pigment = Red, Yellow, Blue (RYB) -- subtractive mixing
24	LASER = monochromatic (one wavelength) + coherent + collimated beam
25	First working laser = **Theodore Maiman (1960)**; used ruby crystal
26	LASER used in: surgery, CD/DVD, barcode scanner, laser cutting, LIDAR
27	Holography = 3D image using laser; invented by **Dennis Gabor (1948)**; Nobel 1971
28	Photoelectric effect = Einstein (1905); Nobel 1921; basis of solar cells, photomultipliers
29	Endoscopy = uses optical fibres to view internal organs without surgery
30	Periscope = uses two plane mirrors at 45 deg; used in submarines